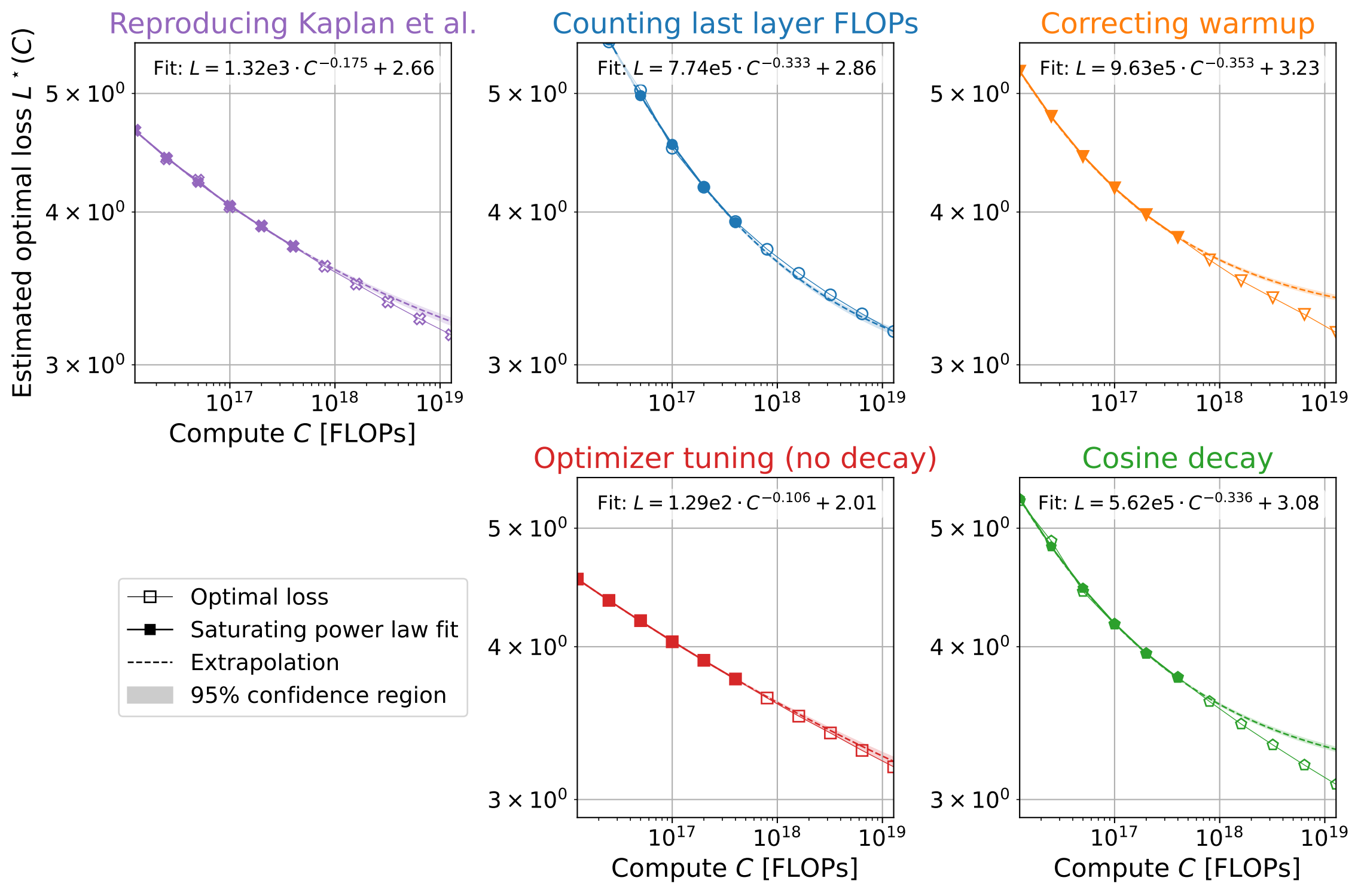
Reproducing Kaplan et al.
Fit: $L = 1.32e3 \cdot C^{-0.175} + 2.66$
Counting last layer FLOPs
Fit: $L = 7.74e5 \cdot C^{-0.333} + 2.86$
Correcting warmup
Fit: $L = 9.63e5 \cdot C^{-0.353} + 3.23$
Optimizer tuning (no decay)
Fit: $L = 1.29e2 \cdot C^{-0.106} + 2.01$
Cosine decay
Fit: $L = 5.62e5 \cdot C^{-0.336} + 3.08$
Estimated optimal loss $L^\star(C)$
Compute C [FLOPs]
5×10^0
4×10^0
3×10^0
10^{17}
10^{18}
10^{19}
Optimal loss
Saturating power law fit
Extrapolation
95% confidence region